TerraSentinel AI

GROUND-MOTION EARLY WARNING FOR INFRASTRUCTURE

Satellites see the ground move. Gemini tells engineers where to go first.

Roads, metro tunnels and dams can sink or slide by a few millimetres for months before anything visible happens. TerraSentinel reads those satellite measurements after every pass, ranks the places that need a person, and uses Gemini on Google Cloud to write an inspection ticket an engineer can act on today.

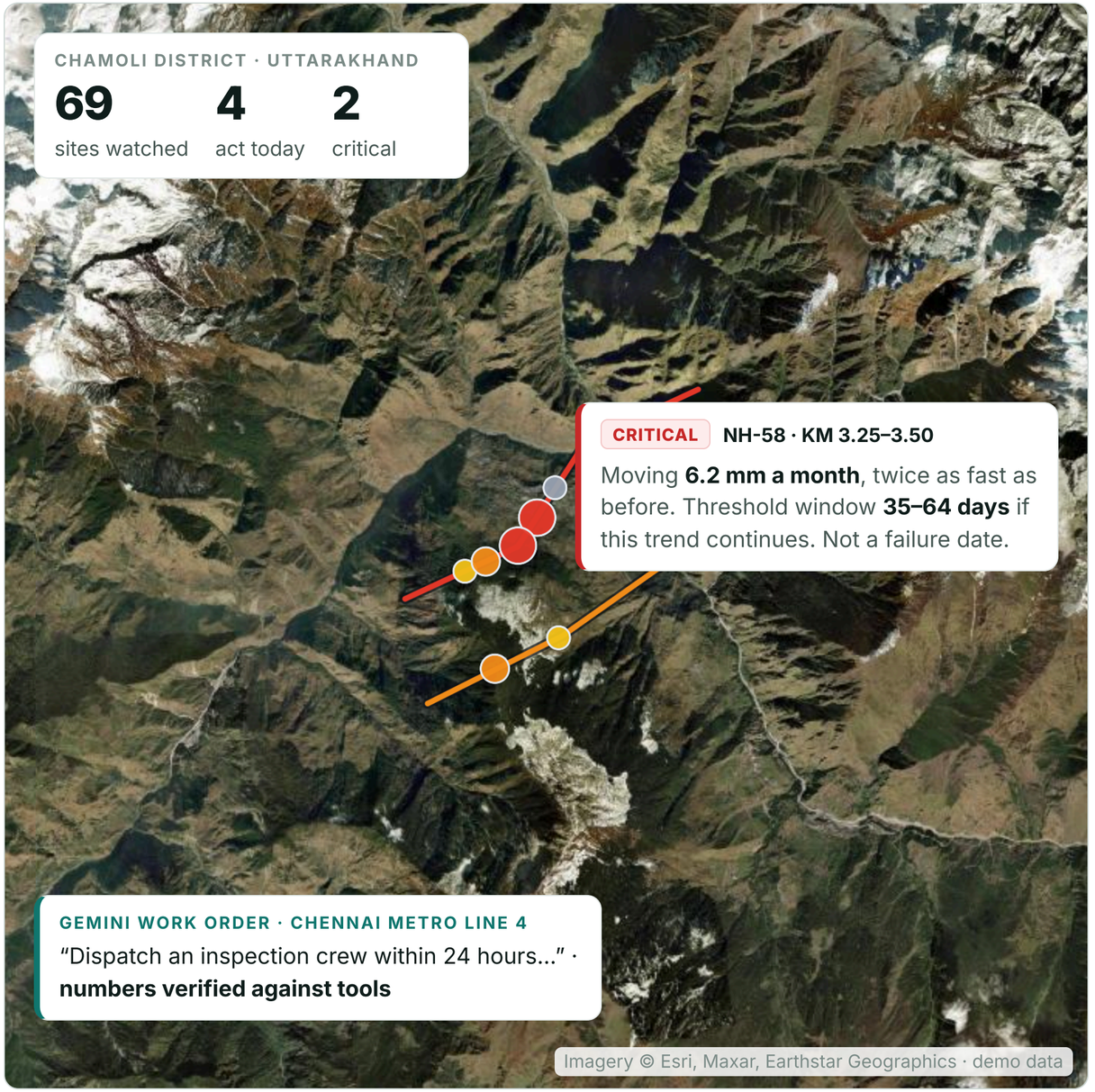
87 → 7
sites to the few that need a crew today

Gemini
writes the ticket, from verified numbers

Google Cloud
Cloud Run · BigQuery · Vertex AI

AR

TEAM
Agent Ram
Ramasamy P · Google Cloud AI Builder Cup 2026 · JAPAC · Sustainability



THE PROBLEM AND THE DECISION WE OWN

Ground creeps for months before anyone can see it. Nobody ranks which creep matters, so crews go to the wrong place.

Mountain highways · NH-58, Uttarakhand
Sole access to the upper valley. One slope slip cuts off towns, pilgrims and supply trucks.

Metro tunnels · Chennai
Settlement above the tunnel drive, under roads and buildings. Tight tolerances, dense exposure.

Dams and tailings · Dhanbad mining belt
Crest settlement and toe movement with severe downstream consequence.

THE ONE DECISION WE SUPPORT

After this satellite pass, which 250 m segment needs an engineer, why, and how soon?

TODAY'S OPTION	WHY IT FAILS THE DECISION
Raw InSAR portal	Thousands of moving points per pass, with no link to assets, no consequence and no "who goes today".
Rainfall alarm	Treats the trigger as the hazard, so it fires over stable ground and buries real creep.
Generic AI chatbot	Writes fluent risk text but invents millimetres and failure dates. Nothing to audit.
Scheduled walk-over	The calendar ignores which segment is accelerating right now.

We do

- Progressive creep and settlement on linear assets
- Rank, explain and track between passes
- Report a threshold-crossing window, with its assumptions

We refuse to claim

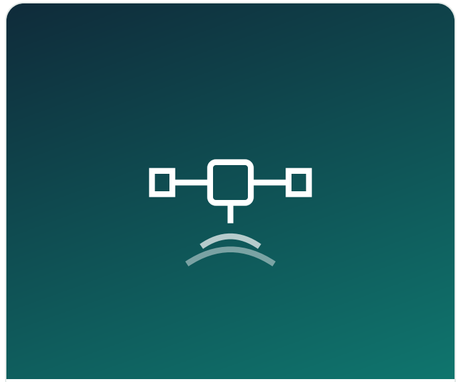
- Failure dates or collapse prediction
- Sudden events between 6–12 day passes
- A certified safety decision

ONE PASS ON THE DEMO DATASET

62,330 displacement observations → **1,355** radar points · 46 passes → **87** 250 m sites on 4 assets → **7** CRITICAL or ELEVATED → **4** "Act today" in Chamoli

THE WHOLE STORY, IN PLAIN WORDS

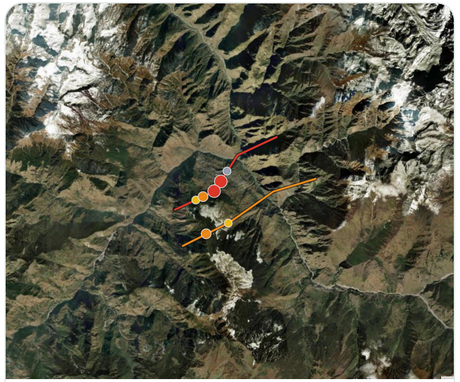
One satellite pass, from space to a crew at the right place.



1 · EVERY 6–12 DAYS

A satellite passes over

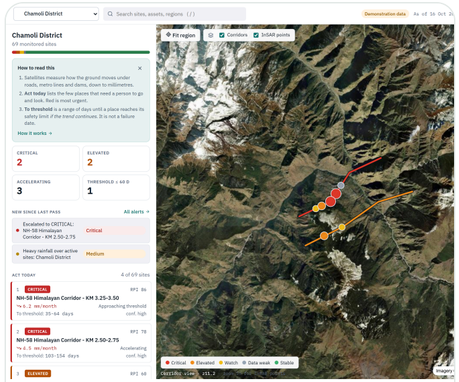
ESA's Sentinel-1 radar measures how much the ground moved since last time, to the millimetre, even through cloud and at night.



2 · AFTER EACH PASS

We map it onto the road

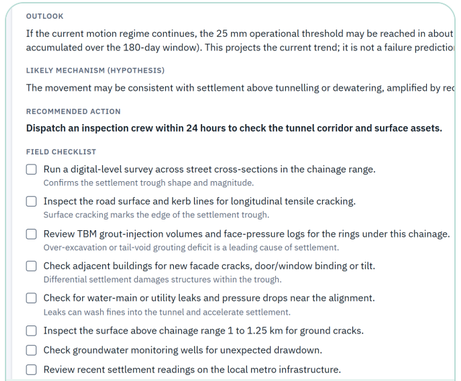
1,355 radar points are matched to the highway, tunnel or dam they sit on and cut into 87 short sections of 250 m.



3 · RANKED

We rank what matters

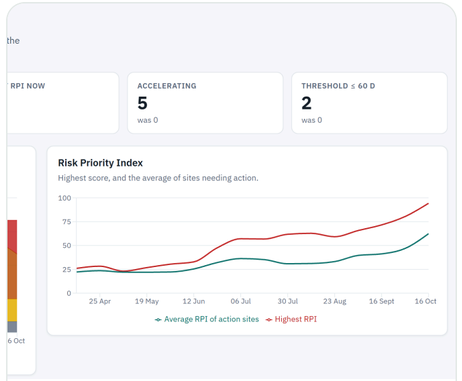
Fast, speeding up, on an important road, with good data? It goes to the top. Rain alone never raises the alarm.



4 · GEMINI

Gemini reasons, then writes the ticket

It reasons over the facts from its tools, summarises what is happening, and creates the inspection ticket. Every number is checked before anyone sees it.



5 · THE NEXT PASS

A crew goes, we keep watching

The engineer works the checklist and records the action. The next pass shows whether the ground calmed down or got worse.

For people who use the road

Problems are found while they are still small, so roads, trains and water supplies are more likely to stay open.

For engineers

A short list instead of thousands of dots, and a ticket that says what to check, where, and why.

For decision-makers

Crews and money go where risk is rising. Every decision can be traced back to the satellite data behind it.

In one line A pass leaves 1,355 moving points. The engine decides which few matter, and **Gemini tells the engineer what to inspect.**

GOOGLE CLOUD AND GENERATIVE AI USAGE

Designed for Google Cloud end to end. Gemini turns a risk score into an action.

INGEST
Cloud Run
Data Hub validates every file and link

→

STORE
BigQuery
GIS join of points to assets, daily history

→

COMPUTE
Cloud Run
tested risk engine scores every site

→

EXPLAIN · GENAI
Gemini · Vertex AI
investigates with tools, writes the work order

→

ACT
Engineers
map, Act today list, ticket, audit trail

Gemini What generative AI does in TerraSentinel

1 Investigates like an analyst
Function calling with 7 read-only tools: metrics, history, outlook, rain, earthquakes, exposure and data quality. The first round must use tools.

2 Writes what people read
Plain-language summary, likely cause (stated as a hypothesis), recommended action with urgency, and a site-specific field checklist.

3 Is checked before it is shown
Every number must match a tool result. Failure-date and certainty language is rejected. On any failure, a labelled template is shown instead.

4 Stays available
Temperature 0.1 and a time budget. Model fallback from `gemini-flash-latest` to lighter models on 429 or 5xx errors.

Live proof: a Gemini ticket drafted during development made 6 tool calls and had 7 numbers verified against them. Its recommendation: *"Dispatch an inspection crew within 24 hours to check the tunnel corridor and surface assets."*

Google Cloud Services and why

Vertex AI	Gemini for production, through the project's own credentials (ADC). AI Studio key for development. **Gemini live today.**
BigQuery	9 tables, partitioned by date and clustered; `ST_DWITHIN` spatial join; views for today's act list and tier history; a cost cap on every query. **Built, one flag to switch on.**
Cloud Run	One container serves the app, API, engine and agent over HTTPS, and scales to zero. **Container ready, public URL next.**
IAM · ADC	Service-account access with no key files, which also satisfies org policies that block key creation.

WITHOUT GENERATIVE AI
An engineer gets a number
"NH-58 KM 3.25–3.50 · RPI 86 · CRITICAL." They still have to dig through charts to work out why, what to check and how soon.

→

WITH GEMINI
An engineer gets a ticket
What is happening, the likely cause, what to do and by when, and a field checklist for that exact site, in plain language.

→

WITH OUR GUARD
The ticket can be trusted
No invented numbers and no promised failure dates. Each ticket carries the tool calls behind it, so anyone can see where every number came from.

DATA: WHAT WE USE, WHERE IT COMES FROM, HOW IT IS CONNECTED

Six sources, one validated path in. Same data model locally or in BigQuery.

SOURCE	WHERE IT COMES FROM	HOW WE CONNECT	ROLE IN THE SCORE
InSAR points + time series	Sentinel-1, processed into persistent scatterers by the owner's InSAR provider	Data Hub upload or HTTPS link · CSV, long or wide (EGMS style)	core measurement
Rainfall	**Open-Meteo** forecast and archive APIs (no key), or owner gauges	`fetch_rain` at each region centre · CSV upload	amplifier only, capped at +50%
Assets	Owner GIS: line, type, criticality 0–1, consequence, buffer	GeoJSON LineString / MultiLineString	consequence
Earthquakes	**USGS** feeds (M2.5+ 30 d) and FDSN (M5+ 5 y), public domain	`public_catalog` via the safe fetcher, cached CSV	context, never scored
Regions · basemap	Country → state → district hierarchy · Esri World Imagery	JSON / CSV · MapLibre raster tiles allow-listed in the CSP	scope and display

1 · CONNECT

`fetch_url`

https only; resolves the host and refuses private and metadata IPs; checks every redirect; size cap and timeout

→

2 · VALIDATE

Check, then import

detects kind and layout; plain-language report; bad rows listed; unsupported formats refused

→

3 · STORE

`TS_DATA_BACKEND`

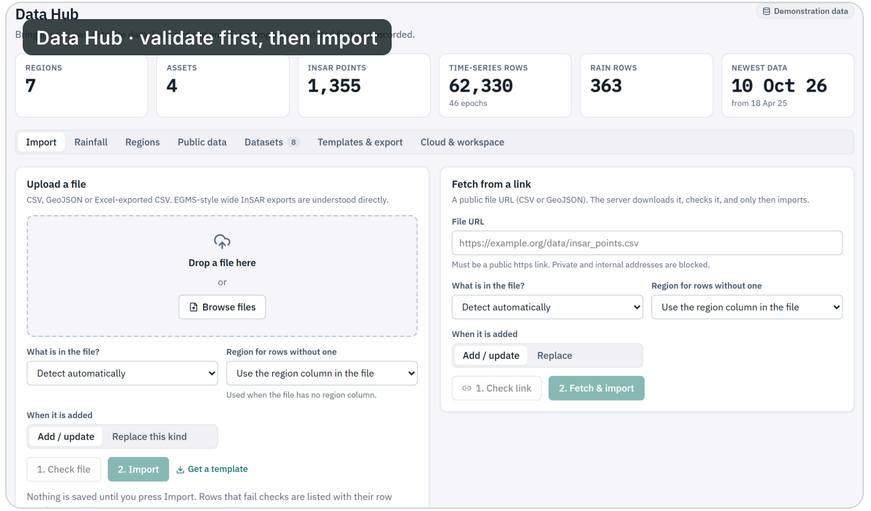
local files or **bigquery** tables; provenance row for every dataset

BIGQUERY LAYOUT

`points · cluster geom, region` `timeseries · partition date` `rain · partition date` `assets · GEOGRAPHY`

`asset_segments · 250 m` `site_snapshots · partition as_of` `alert_actions · audit` `datasets · provenance` `regions`

Each job sets `maximum_bytes_billed`. BigQuery uses Google Cloud credentials (ADC); the Gemini key is never a BigQuery credential.



THE RISK ENGINE · DETERMINISTIC, TESTED, EXPLAINABLE

Measure, score, project. Plain rules first, then the exact maths.

1 · MEASURE

How fast is it moving, and is it speeding up?

Recent speed	slope over the last **60 days**, mm/month
Baseline	slope from **270 to 90 days ago**
Acceleration	|recent| ÷ max(|baseline|, 0.5)
Persistence	steps in the same direction, smoothed
Seasonal	a big swing that mostly recovers
Site series	median of the points in each 250 m section

CAN WE TRUST THE DATA?

```
conf = .30·coh + .15·epochs + .20·points + .20·agree + .15·SNR
```

Capped at 0.30 for low coherence or fewer than 3 points, so weak data can never be CRITICAL.

2 · SCORE

Moving × important × trustworthy

```
kin = 100·(.4speed + .3accel + .3progress)
hazard = kin·(1 + 0.5·rain_14d/200 mm)
RPI = hazard × (0.5+0.5·criticality)
      × (0.5+0.5·confidence)
```

speed = |recent| ÷ 5·creep · accel = (ratio−1) ÷ 3 · progress = window ÷ threshold · all clipped 0–1

- Slow and far from the limit: × **0.2**
- Seasonal swing: × **0.35**
- Weak data: capped at **15**
- Earthquakes: shown, **never scored**

CRITICAL	near the limit, or speeding up on an important asset with heavy rain
ELEVATED	speeding up, or steady creep on an important asset
WATCH	seasonal, or creep on a lesser asset

3 · PROJECT

If nothing changes, how long until the limit?

```
ETA_const = remaining ÷ speed
ETA_accel = root of a·t²+b·t+c = T
band = [min, max] ≤ 365 d
close? → [.75, 1.35]·mid
```

ASSET	DEMO LIMIT	CREEP
Highway	40 mm / 180 d	2.0 mm/mo
Metro	25 mm / 180 d	1.5 mm/mo
Dam	30 mm / 180 d	1.0 mm/mo

Always a band, never a date. The 0.4 / 0.3 / 0.3 weights, the rain cap and these limits are demo defaults. The asset owner replaces them with their own standard. Assumes the trend continues.

WHAT-IF ON THE NH-58 SITE

- **As observed** → RPI **86** CRITICAL
- **Minor road (0.3)** → **57** ELEVATED
- **Poor radar data** → **≤ 15** WEAK
- **Heavy rain, stable** → **≈ 1** STABLE

ONE SITE, END TO END, AS THE PRODUCT SHOWS IT

NH-58 KM 3.25–3.50: CRITICAL, RPI 86. Threshold window 35–64 days if this trend continues.

NH-58 Himalayan Corridor - KM 3.25-3.50

CRITICAL Approaching threshold KM 3.25-3.50

Now | Trend | History | Outlook | Action

Cumulative displacement 29.7 mm of 40 mm operational threshold; projected crossing in 35-64 days.

Previous pass (28 Sept 26): CRITICAL RPI 61

RECENT SPEED	BASELINE SPEED
6.2 mm/month last 60 days	2.8 mm/month earlier regime
ACCELERATION	**LAST 60 DAYS**
2.2x alert at ≥ 1.8x	12.4 mm cumulative

TOWARD OPERATIONAL THRESHOLD 29.7 mm / 40 mm

over a 180-day window · projected crossing: 35-64 days

EXPOSURE

NH-58 Himalayan Corridor (demo) · criticality 0.95
Sole arterial access to the upper valley; high daily transit volume.

TRIGGER CONTEXT

213 mm rain in the last 14 days (peak day 25 mm). Rain amplifies existing motion; it is not a hazard on its own. Source: DEMO-SYNTHETIC.

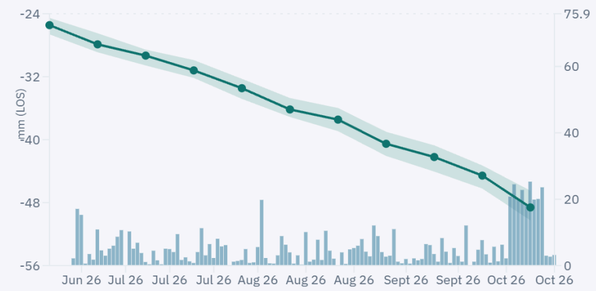
EARTHQUAKES NEARBY (CONTEXT)

2 earthquakes of M4+ within 100 km in the last 30 days, largest M5.1, nearest 57 km away.
Largest in the stored history within 200 km: M5.7, 39 km NE of Dipayal, Nepal, 03 Oct 23.
Context only: earthquakes are not used in the risk score. Source: U.S. Geological Survey, Earthquake Hazards Program (earthquake.usgs.gov).

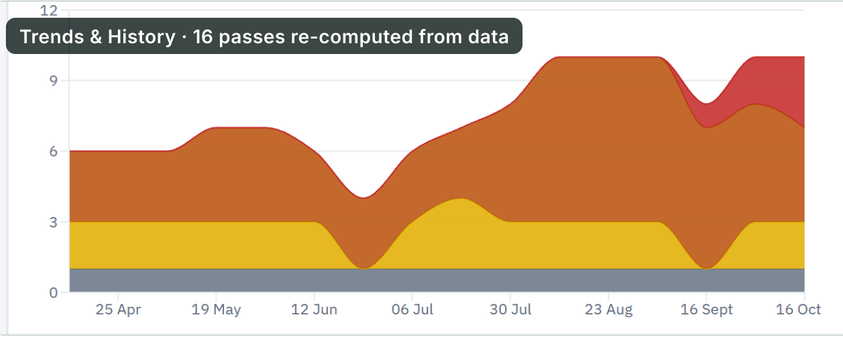
WHY IT RANKS HERE

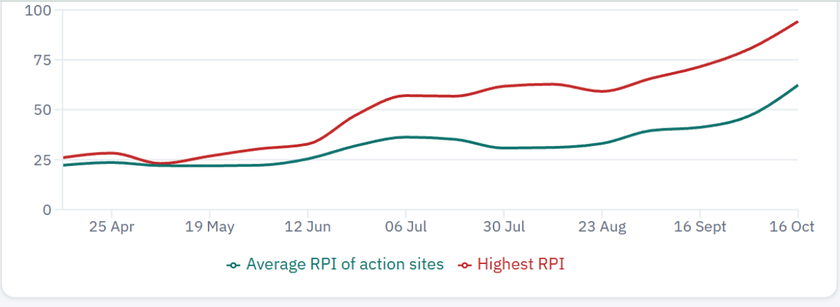
Kinematic score	59.4
Rain trigger	100
Hazard (rain amplifies motion)	89
Consequence factor	x 0.97
Confidence factor	x 0.99
Risk Priority Index	**85.7**

STEP	CALCULATION	RESULT
Inputs	6.2 vs 2.8 mm/mo · 2.2× · 29.7/40 mm · 213 mm rain · crit 0.95 · conf 0.97	
Kinematic	100 × (.4·0.62 + .3·0.40 + .3·0.74)	**59.4**
Hazard	59.4 × (1 + 0.5 × 1.0) · rain capped	**89.1**
RPI	89.1 × 0.975 × 0.985	**85.7**
Tier	threshold approach · crit ≥ 0.5	CRITICAL
Outlook	51 d constant · 44 d accelerating · widened · not a failure date	**35–64 d**



Last 120 days. Bars show daily rainfall; the shaded band is the spread across scatterers at this site.





61 → 86
this site's RPI since the last pass

3 → 7
action sites over 16 passes

M5.1
largest nearby quake: context only

same
numbers in UI, tools and work order

GENERATIVE AI, ENGINEERED FOR TRUST

Gemini investigates with tools. A guard verifies every number before an engineer sees it.

TS-IN-TN-CHN-METRO4-DEMO-004-20261016

Live ticket · 6 tool calls · 7 numbers verified

CRITICAL **TIER 1 - URGENT FIELD INVESTIGATION**

Urban Metro Line 4 Tunnel Corridor - Chennai (demo) KM 1.00 - 1.25

13.08362, 80.27419 · as of 2026-10-16

Gemini (gemini-flash-lite-latest) · Numbers verified against tools (7) · 6 tool calls

GROUNDED BRIEF

Site METRO4-DEMO-004 is at a critical state with high confidence data. Cumulative displacement is 18.45 mm towards a 25 mm threshold, with an estimated time-to-threshold of 23 to 41 days if the current motion continues.

EVIDENCE

Recent speed	6.4 mm/month (baseline 1.0 mm/month; 6.6x)	get_site_metrics

OUTLOOK

If the current motion regime continues, the 25 mm operational threshold may be reached in about 23-41 days (18.4 mm of 25 mm accumulated over the 180-day window). This projects the current trend; it is not a failure prediction.

LIKELY MECHANISM (HYPOTHESIS)

The movement may be consistent with settlement above tunnelling or dewatering, amplified by recent rainfall of 72.3 mm over 14 days.

RECOMMENDED ACTION

Dispatch an inspection crew within 24 hours to check the tunnel corridor and surface assets.

FIELD CHECKLIST

- [ ] Run a digital-level survey across street cross-sections in the chainage range.
  Confirms the settlement trough shape and magnitude.
- [ ] Inspect the road surface and kerb lines for longitudinal tensile cracking.
  Surface cracking marks the edge of the settlement trough.
- [ ] Review TBM grout-injection volumes and face-pressure logs for the rings under this chainage.
  Over-excavation or tail-void grouting deficit is a leading cause of settlement.
- [ ] Check adjacent buildings for new facade cracks, door/window binding or tilt.
  Differential settlement damages structures within the trough.
- [ ] Check for water-main or utility leaks and pressure drops n
  Leaks can wash fines into the tunnel and accelerate settlement.

Written by Gemini · cause as a hypothesis · action · field checklist

ROUND 0 **ANY** must call tools → 1–2 **AUTO** investigate → ROUND 3 **NONE** write JSON → GUARD **Verify** pass or template

7 READ-ONLY TOOLS, THE ONLY SOURCE OF NUMBERS

`get_site_metrics` `get_displacement_history` `get_outlook` `get_rainfall_context` `get_seismic_context` `get_asset_exposure` `get_data_quality`

The guard rejects

- any mm, days, ×, %, km or point count not in a tool result
- “will collapse”, “guaranteed”, “failure date”, “no risk”
- unhedged causes; weak data without a warning

“...an estimated time-to-threshold of **23 to 41 days** if the current motion continues.”
passed: quoted from the live ticket, values from get_outlook

“...the tunnel will fail in **21 days**.”
rejected: unknown number and failure language

Resilience: the model chain falls back from `gemini-flash-latest` on 429 or 5xx errors. This ticket came from `flash-lite`. Any failure still ships a labelled template.

One container, two storage back-ends, the same tested maths.

Edge	**Cloud Run** · HTTPS · CSP and HSTS · rate limits · session cookie
UI	**React 19** · MapLibre satellite map · React Query · role-aware screens
API	**FastAPI** · RBAC viewer / analyst / admin · CSRF · SSRF-safe fetch · audit log
Engine	**numpy · pandas · shapely** · metrics, RPI, tiers, bands, alerts, pass replay
GenAI	**Gemini** · tool loop · number guard · model fallback · AI Studio or Vertex AI
Data	**BigQuery** (GEOGRAPHY, partitions, snapshots) or local · USGS store · provenance

Ingest validate → **Store** BigQuery → **Compute** engine → **Snapshot** history → **Serve** UI + Gemini

Engineering invariants

- GenAI never computes risk, tier or outlook
- History is re-run from data, so every past score is reproducible
- Public data is context only and never moves a score
- Every dataset has provenance; every alert action is audited
- BigQuery jobs carry a bytes-billed cap; credentials come from ADC, never key files

VERIFIED

349 backend tests · security controls · UI on a running server · live USGS fetch · **live Gemini work order**

NEXT

Cloud Run URL · BigQuery on a real project · partner InSAR · OIDC / IAP sign-in · field validation study

Every ranked site comes with its formula, its data and its limits.

Buyer: the asset-integrity or geotechnical team at a highway, metro, dam or mine. They buy a shorter list: 1,355 points become 7 sites. The same rank, explain and inspect loop fits any linear asset.

Ask: review the live demo, the 3-minute video and the repository.

Team Agent Ram · Ramasamy P

DEPLOYMENT ON GOOGLE CLOUD

Engineers browser · role → **CLOUD RUN SERVICE · ONE IMAGE** (`React SPA` · `FastAPI` · `detection engine` · `Gemini agent` · service account via ADC · no key files) → BigQuery · dataset + views; Gemini · AI Studio / Vertex AI; Open-Meteo · USGS · Esri (safe egress)

SECURITY AND OPERATIONS

scrypt passwords · hashed sessions · RBAC · CSRF + Origin check · SSRF guard · CSP · HSTS · rate limits · login throttling · request IDs · audit log · BQ doctor

TerraSentinel AI

Thank you.

Which 250 metres should an engineer inspect today? The engine ranks them. Gemini writes the ticket, and every number is checked.

Gemini · Vertex AI | BigQuery | Cloud Run | Sentinel-1 InSAR

WHO PAYS
Asset-integrity and geotechnical teams at highway, metro, dam and mine operators.

WHAT THEY BUY
Engineer attention. 1,355 radar points become 7 sites to review.

Same loop on any linear asset: rank, explain, inspect, record, next pass. Ask: review the live demo, the 3-minute video and the repository.

TEAM

Agent Ram

NAME
Ramasamy P

EMAIL
ramp21052000@gmail.com

PHONE
+91 90250 32130

Google Cloud AI Builder Cup 2026 · JAPAC · Sustainability